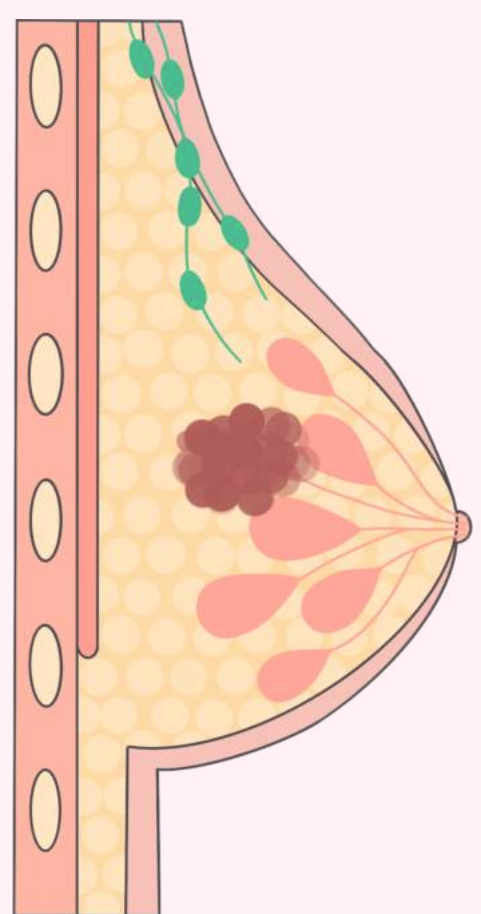


MT1 Melatonin Receptor Signaling and ERK-Mediated Stress Responses in MCF-7 Cells

Background



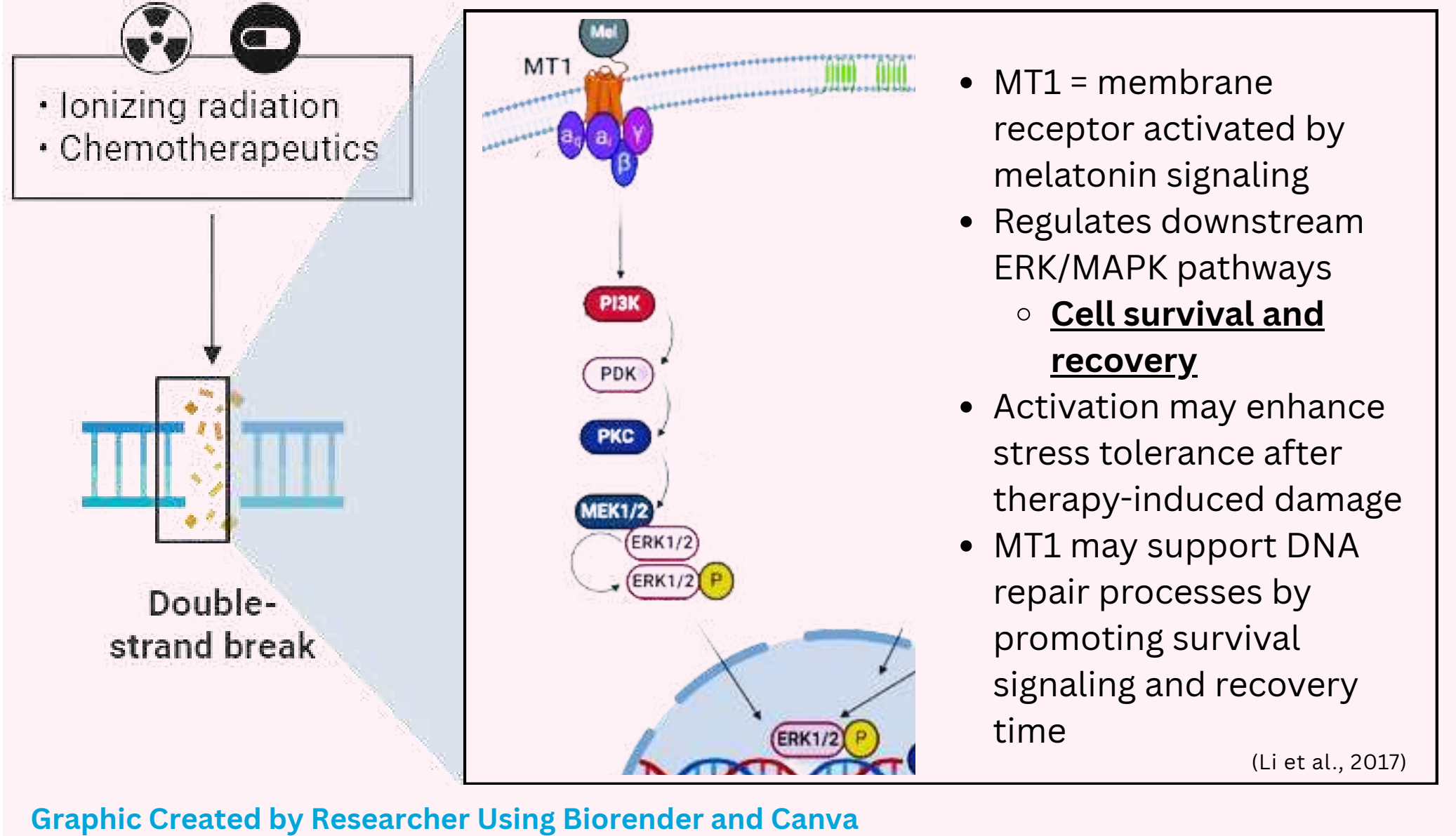
Breast Cancer

- **Most common cancer** among women worldwide (Giaquinto et al., 2024)
- Standard therapies include **chemotherapy, radiation**, and targeted treatment
- Many treatments work by generating oxidative stress or **inducing DNA damage**
- Tumor **resistance to damage-response** therapy remains a major clinical challenge

Melatonin (Li et al., 2017)

- Hormone produced by the pineal gland
- **Commonly used by cancer patients** as an over-the-counter sleep supplement
- Antioxidant activity, cell signaling, apoptosis, and cellular stress responses
- Interacts with cancer-related pathways through membrane **receptors such as MT1**





Significance

- Therapy resistance remains a major barrier in breast cancer treatment, as many tumor cells survive oxidative and DNA-damaging stress.
- Melatonin signaling through the MT1 receptor may activate protective pathways that help cancer cells recover after damage.
- Clarify role of MT1 in cell survival and p-ERK expression post-oxidative stress induction

Project Overview

- Luzindole is a melatonin receptor antagonist used to block MT1 signaling.
- Hydrogen peroxide (H₂O₂) was used to model oxidative stress, which can generate reactive oxygen species and induce DNA strand damage.
- This study tested whether MT1 inhibition with luzindole changes MCF-7 breast cancer cell survival during oxidative stress exposure.

Hypothesis

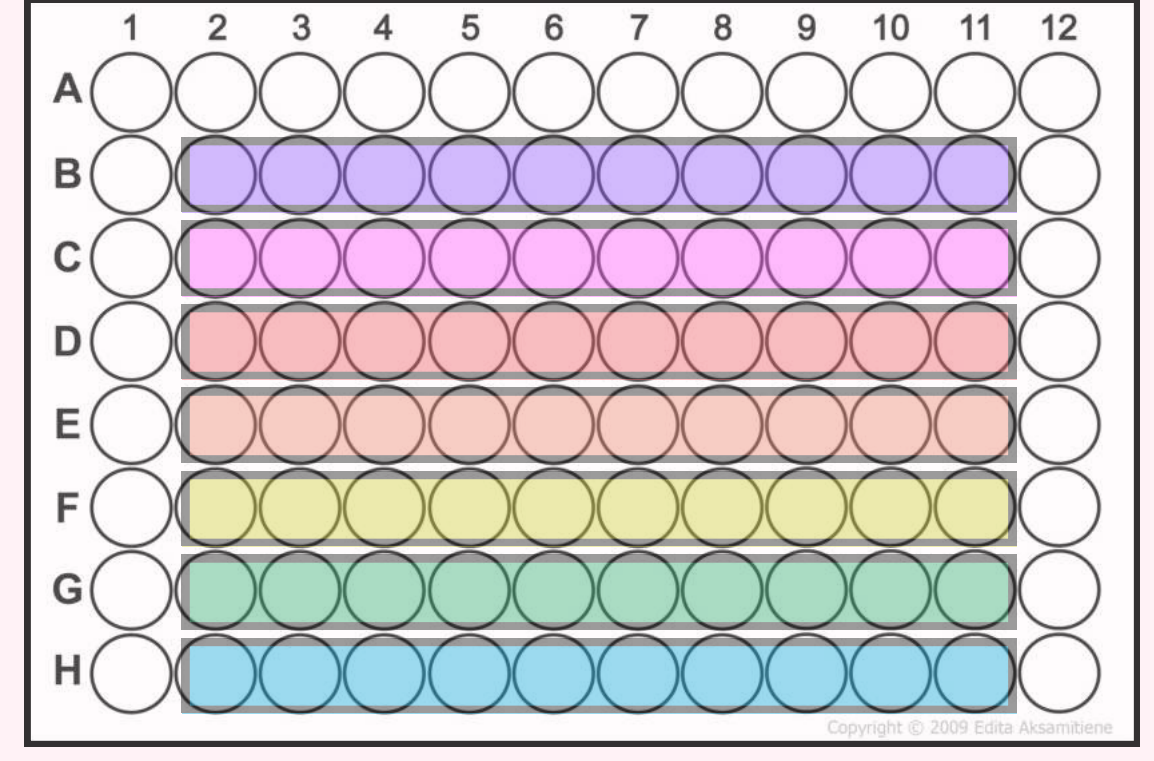
If: MCF-7 cells are treated with Luzindole to block MT1 receptor signaling

Then: the expression of p-ERK will increase, thereby increasing cell survival under oxidative stress conditions

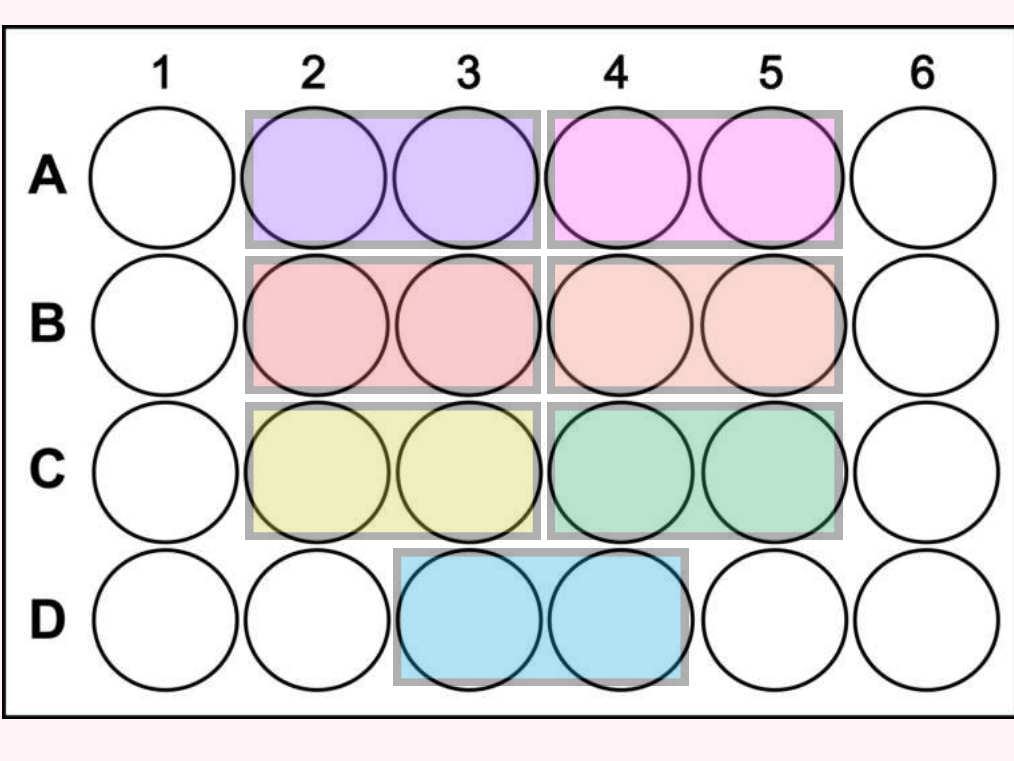
Justification: Blocking MT1 signaling removes melatonin-mediated regulation of MAPK signaling, which can lead to compensatory activation of ERK phosphorylation and enhanced downstream pro-survival signaling in cancer cells.

Methodology

1. Seed Well Plates



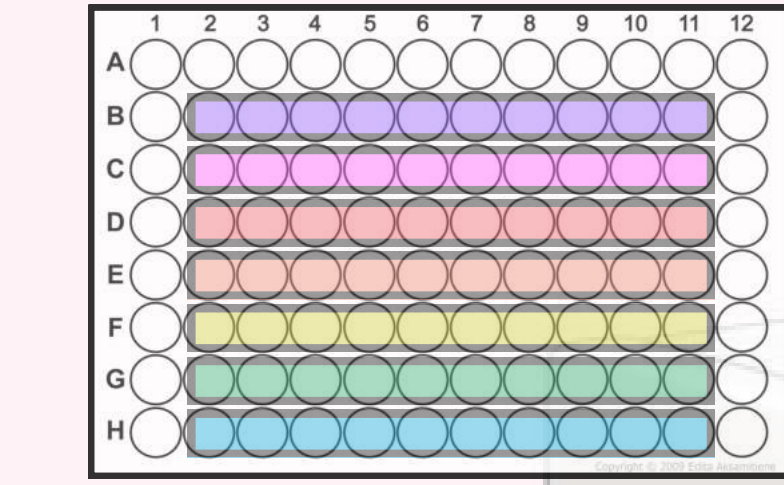
2. Administer Treatments



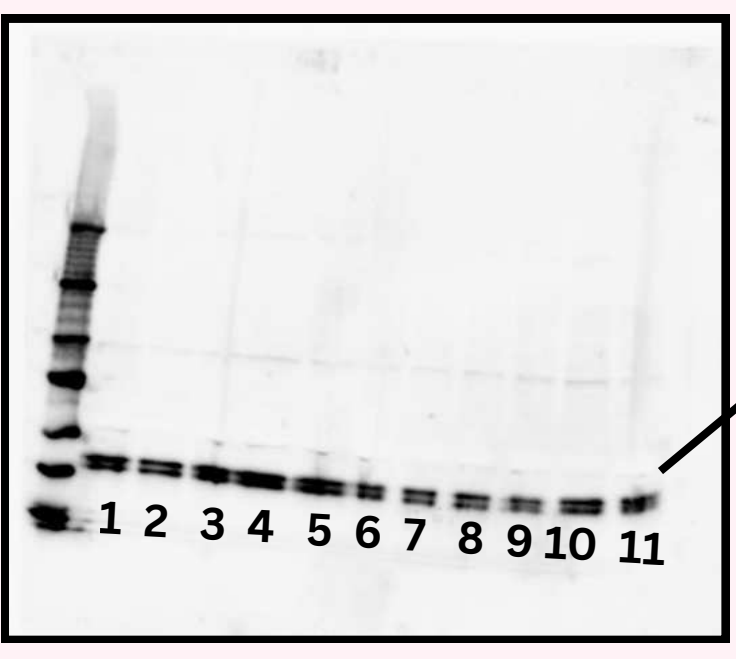
- Negative Control:** Media Only
- Vehicle Control:** Media + DMSO
- 10 uM Luzindole + Media**
- 10uM Luzindole + H₂O₂ + Media**
- 30uM Luzindole + Media**
- 30uM Luzindole + H₂O₂ + Media**
- H₂O₂ Only + Media**

Independent Variable: Luzindole Concentration
H₂O₂ used to induce oxidative stress

3. Acid Phosphatase Assay



4. Western Blot



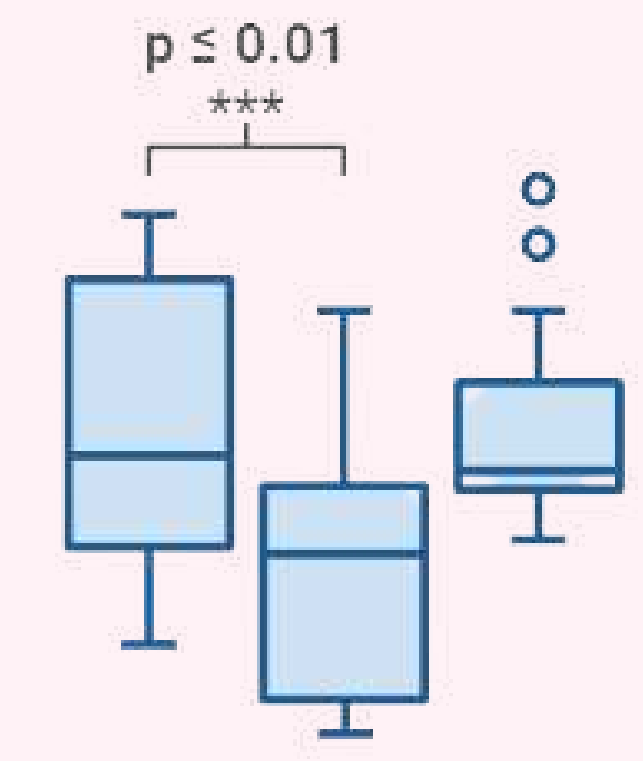
chemiluminescence reaction

secondary anti-rabbit

primary Anti-ERK antibody

ERK

5. Statistical Analysis



p ≤ 0.01

Parametric ANOVA tests used for statistical analysis

Results

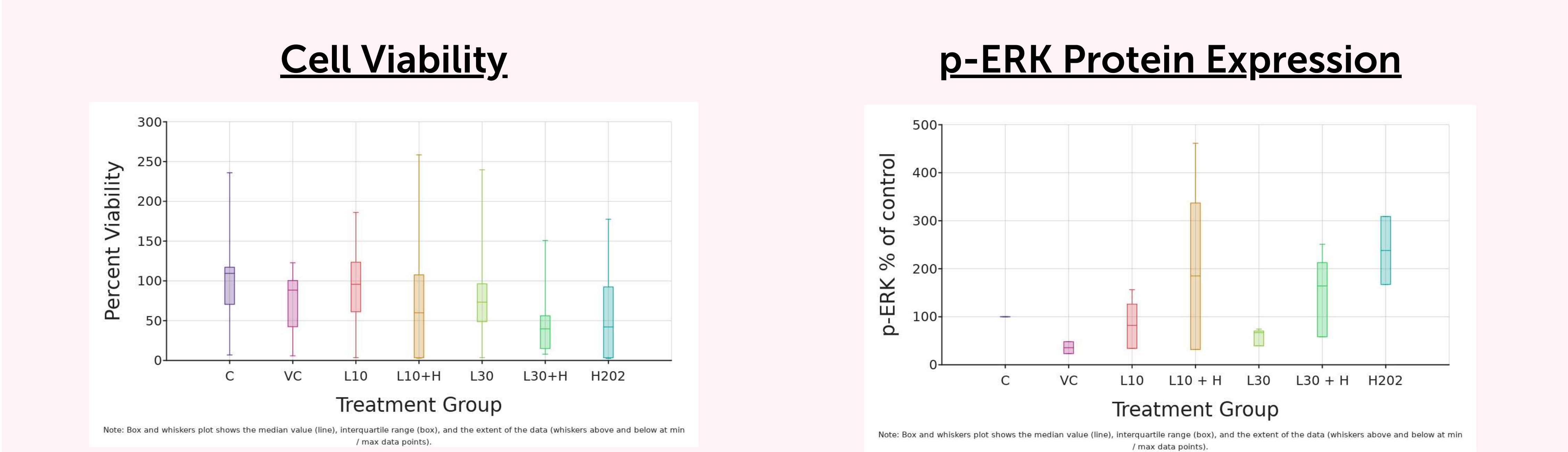


Figure 02. Percent Viability (relative to control) compared by all treatment groups as measured by Acid Phosphatase Assay

Legend: C - Control VC - Vehicle Control L10 - Luzindole 10 µl L10 + H - Luzindole 10 µl + H₂O₂ L30 - Luzindole 30 µl L30 + H - Luzindole 10 µl + H₂O₂ H2O2 - H₂O₂ only

ANOVA Results	
Groups With Significant Differences	P-Value
L30+H - C	<0.01
H ₂ O ₂ - C	0.01
L30+H - L10	0.01

A P-value of <0.01 means extremely strong confidence that the groups are different.

Table 1. Significant ANOVA p-values comparing cell viability across different treatment and control groups.

ANOVA Results	
Effect	Degrees of Freedom
Treatment Group	6
	F-statistic
	1.3
	P-value
	0.3

A p-Value of 0.3 means little to no indication that the groups might be different

Table 2. ANOVA statistical analysis on p-ERK expression across different treatment groups

Discussion

Data collected thus far **did not support hypothesis**. Statistical analysis concluded that there was not enough evidence to support a difference in p-ERK expression, and contrary to hypothesis, viability was seen to decrease upon addition of Luzindole.

- Acid Phosphatase for Cell Viability**
- Greatest viability reduction in 30 µM luzindole + H₂O₂ groups → impaired oxidative stress tolerance under stronger MT1 inhibition
 - **Higher viability in H₂O₂-only vs. 30 µM luzindole + H₂O₂ conditions** → protective role of MT1-associated signaling during oxidative stress
 - Lower-dose luzindole = more viability post-oxidative stress→ possible compensatory/adaptive signaling response
 - dose-dependent response

- Western Blot for p-ERK expression**
- **No statistically significant differences between treatment groups (p = 0.30)**
 - Oxidative stress and MT1 inhibition altered viability without producing major detectable changes in ERK phosphorylation
 - Suggests p-ERK signaling **may not be the primary pathway** driving survival differences in MCF-7 cells under these conditions
 - Although minor upward trends appeared in oxidative stress groups, variability between samples prevented conclusive evidence of altered ERK activation
- Conclusion:**
- Oxidative stress-induced viability changes **occurred largely independently** of significant p-ERK expression changes

- Implications**
- Significant decreases in viability + non-significant p-ERK differences → MT1 signaling may regulate oxidative stress responses through pathways other than ERK
 - Blocking MT1 receptors **increased susceptibility** to oxidative damage, particularly at higher luzindole concentrations, **without substantially altering ERK phosphorylation**
 - Indicates ERK signaling alone may be insufficient to explain survival outcomes in MCF-7 cells during oxidative stress

Next Steps

- Investigate additional oxidative stress-response pathways beyond ERK
 - AKT, p38 MAPK, apoptotic signaling
- Perform time-course western blot analysis
 - capture transient changes in ERK phosphorylation
- Test additional luzindole concentrations
 - evaluate dose-dependent MT1 inhibition effects
- Conduct melatonin rescue experiments
 - determine whether melatonin reverses reduced viability under oxidative stress conditions

Key References

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